RestaurantOS Version 1.0 - Product Requirements Document (PRD)

Version: 1.0
Project Type: Multi-Tenant SaaS Platform
Target Customers: Small and Medium Restaurants
Goal: Build a production-ready QR-based restaurant ordering platform that restaurants can start using immediately.

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1. Vision

RestaurantOS is a cloud-based software platform that allows restaurants to receive customer orders digitally through QR codes placed on tables.

Customers can browse the menu, place orders, make online payments, and track the order status.

Restaurant staff manage incoming orders through a web dashboard.

RestaurantOS is designed to simplify restaurant operations without forcing restaurants to completely change their existing workflow.

The first version intentionally focuses only on ordering, payments, and menu management.

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2. Objectives

RestaurantOS Version 1.0 should allow a restaurant to:

- Receive customer orders digitally.
- Eliminate printed menus.
- Accept online payments.
- Manage menu items.
- Update order status.
- Generate QR codes for tables.
- Manage everything through one simple web dashboard.

The system should be simple enough for any restaurant owner to start using within one hour of setup.

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3. User Types

There are only three user types.

Customer

Customers interact only through QR codes.

No account creation is required.

Restaurant

Restaurant owners and managers log into the admin dashboard.

Kitchen staff and waiters are **not** given separate interfaces in Version 1.0.

Restaurant staff continue their normal workflow.

The manager updates order status manually.

Platform Administrator

The platform owner (RestaurantOS) manages all restaurants using an Admin Dashboard.

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4. Customer Flow

Step 1

Customer sits at a table.

Every table has a unique QR code.

Example:

Table 12

Restaurant ABC

Step 2

Customer scans QR.

Website opens automatically.

No application download.

No login.

Step 3

Customer enters

- Name
- Mobile Number

The table number is automatically identified from the QR code and cannot be edited.

Step 4

Customer views the restaurant menu.

Menu contains:

- Categories
- Dish Image
- Dish Name
- Description
- Price
- Availability

Unavailable dishes are hidden or clearly marked.

Step 5

Customer adds dishes to cart.

Customer can

- Increase quantity
- Decrease quantity

- Remove item

Step 6

Customer proceeds to Checkout.

Checkout page displays:

- Ordered Items
- Quantity
- Individual Prices
- Taxes
- Grand Total

Step 7

Customer selects payment method.

Version 1.0 supports:

- UPI
- Cards
- Net Banking
- Wallets

(through a payment gateway)

Step 8

Payment succeeds.

Order is automatically created.

Restaurant receives the order instantly.

Step 9

Customer is redirected to Order Tracking.

Three statuses exist.

- Order Received
- Preparing
- Served

The page automatically updates.

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5. Restaurant Dashboard

After login, restaurant owners access the dashboard.

Sidebar contains:

- Dashboard
- Orders
- Menu
- Tables
- Settings
- Profile

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6. Dashboard

Dashboard displays:

Today's Revenue

Today's Orders

Pending Orders

Preparing Orders

Completed Orders

Total Sales Today

Recent Orders

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7. Order Management

Restaurant receives orders immediately after successful payment.

Each order contains:

Order Number

Date

Time

Table Number

Customer Name

Customer Mobile Number

Payment Status

Payment Method

Items Ordered

Quantity

Item Notes (future)

Total Amount

Current Status

Assigned Waiter (optional)

Actions:

- Change Status to Order Received
- Change Status to Preparing
- Change Status to Served

Restaurant may optionally assign a waiter for internal tracking.

Kitchen communication remains manual.

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8. Menu Management

Restaurant can

Add Dish

Edit Dish

Delete Dish

Mark Dish Available

Mark Dish Unavailable

Upload Dish Image

Edit Description

Edit Price

Create Categories

Reorder Categories

Each dish contains:

- Name
- Description
- Price
- Category
- Image
- Availability

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9. Table Management

Restaurant can

Create Table

Delete Table

Rename Table

Generate QR Code

Download QR Code

Print QR Code

Each QR uniquely identifies:

Restaurant

Table Number

Customer should never manually enter the table number.

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10. Restaurant Settings

Restaurant can manage:

Restaurant Name

Logo

Address

Phone Number

GST Number (optional)

Business Hours

Accepted Payment Methods

Currency

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11. Notifications

Restaurant Dashboard

New Order Notification

Customer

Order Status Updates

No SMS or WhatsApp notifications in Version 1.0.

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12. Platform Admin Dashboard

Accessible only by RestaurantOS administrators.

Features:

Dashboard

Restaurant Management

Subscriptions

Support

Restaurant Management

Create Restaurant

Suspend Restaurant

Activate Restaurant

Delete Restaurant

Reset Restaurant Password

View Restaurant Details

Dashboard

Total Restaurants

Active Restaurants

Suspended Restaurants

Today's Orders Across Platform

Revenue Overview

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13. Authentication

Restaurant Login

Platform Admin Login

Customers do not require authentication.

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14. Payment System

Payment gateway integration is mandatory.

Only successful payments create orders.

Failed payments do not create orders.

Every payment stores:

Transaction ID

Amount

Gateway Response

Timestamp

Payment Status

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15. Database Structure (High-Level)

Restaurant

Users (Restaurant/Admin)

Tables

Menu Categories

Menu Items

Orders

Order Items

Payments

Restaurant Settings

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16. Technical Requirements

Architecture:

Multi-Tenant SaaS

Every restaurant has isolated data.

One shared application serves all restaurants.

Responsive Design

Desktop

Tablet

Mobile

Modern UI

Fast Performance

Secure Authentication

Role-Based Access

Automatic Error Logging

Cloud Deployment

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17. Version 1.0 Exclusions

The following features are intentionally excluded:

- Kitchen Dashboard
- Waiter Dashboard
- Inventory Management
- Employee Attendance
- Payroll
- Customer Accounts
- Loyalty Program
- Coupons
- Discounts
- Reviews
- Table Reservations
- Delivery Management
- Multi-Branch Management
- AI Features
- WhatsApp Notifications
- SMS Notifications
- Kitchen Printer Integration
- POS Hardware Integration
- Offline Mode
- Multi-Language Support

These features are reserved for future versions.

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18. Success Criteria

RestaurantOS Version 1.0 is considered successful when the following workflow works reliably:

1. Platform admin creates a restaurant account.
2. Restaurant logs in.
3. Restaurant uploads its menu.
4. Restaurant creates tables.
5. QR codes are generated.
6. Customer scans the QR.
7. Customer enters name and mobile number.
8. Customer browses the menu.
9. Customer adds items to the cart.
10. Customer completes online payment.
11. Order instantly appears in the restaurant dashboard.
12. Restaurant updates the order status.
13. Customer sees the updated status in real time.

If all of these steps function smoothly, RestaurantOS Version 1.0 is complete and ready for pilot deployment with real restaurants.

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19. Long-Term Vision

RestaurantOS is not just a QR ordering platform. Version 1.0 lays the foundation for a complete restaurant operating system.

Future versions may introduce:

- Kitchen Dashboard
- Waiter Dashboard
- Inventory Management
- Loyalty Programs
- Table Reservations
- Multi-Branch Support
- Customer Analytics
- AI Sales Insights
- AI Demand Forecasting
- WhatsApp Integration
- Advanced Reporting
- POS Integrations
- Restaurant Mobile Apps

The guiding principle is to keep Version 1.0 focused, reliable, and easy to adopt while leaving room for future expansion.